Prince Baid

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.93
Delhi Public School, Pal, Jodhpur	Class XII (CBSE)	2024	95.60%
Delhi Public School, Pal, Jodhpur	Class X (CBSE)	2022	98.40%

PROJECTS & RESEARCH EXPERIENCE

Traffic Flow Modeling & Transportation Network Optimization | Under Prof. Digvijay S. Panwar, IITH

- Developed graph-based models and Python simulations for traffic flow, congestion formation, and shockwave propagation using classical traffic flow theories on large-scale urban transportation networks.
- Implemented **BFS, DFS & Dijkstra's** in **C++** for routing, shortest-path computation and transport-network analysis.
- Investigated graph-theoretic traffic modeling, developed a **Random Forest-based congestion prediction model** using traffic and network features, and explored **LWR/PDE-based traffic analysis**, using visualization and analytics tools to evaluate network performance under dynamic traffic conditions.

Probabilistic Global Optimization (ProGO) | Research Assistant, Prof. Soumyajit Ghosh, NUS

- Studied and implemented **ProGO**, a probabilistic framework for gradient-free global optimization that constructs distributions increasingly concentrated around global optima; implemented the **Latent Slice Sampler (LSS)** to sample from the nascent-minima distribution and investigated its concentration around global optima.
- **Reproduced and computationally validated the 1-D formulation**, analyzing the evolution of the minima distribution with the concentration parameter.
- **Benchmarked ProGO against gradient-based, zeroth-order, and Bayesian optimization methods** across non-convex test functions up to **1000 dimensions**, evaluating log-regret and convergence; observed substantially lower regret on high-dimensional Ackley/Levy benchmarks, e.g. **Ackley d=20: function/minima log-regret = 35.35/35.86**

FinSecure AI (Credit Risk Assessment Model) :

- Developed an **ML pipeline** for credit risk assessment using **Python, Pandas, NumPy, and Scikit-learn**, performing data preprocessing, feature engineering, and model evaluation on structured financial datasets.
- Built and compared **Logistic Regression, Decision Tree, and Random Forest** models, and designed the technical architecture of an **AI-assisted credit scoring framework** for the **FinShield Hackathon**.

SKILLS

- **Programming Languages:** C/ C++ , Python, SQL
- **Software Tools, and Libraries:** NumPy, Pandas, Scikit-learn, PyTorch, Matplotlib, Git, GitHub
- **Concepts and Technologies:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS
- **Data Analytics & Machine Learning:** Regression, Classification, Clustering, ML Algorithms, Data Preprocessing, Feature Engineering, Anomaly Detection, Time-Series Analysis, Model Evaluation, Hyperparameter Tuning, Cross Validation
- **Relevant Courses:** Probability (A+), Statistics (A)**,** Linear Algebra, Complex Variables (A-)**,** Differential Equations (A-), Calculus, Data Structures & Applications
- **Soft Skills:** Communication, Team Management, Critical Thinking, Problem Solving, Leadership

ACHIEVEMENTS

- **Top 1000 Finalist, Youth Ideathon India 2023** -- Designed, and developed **HydroBella**, an innovative rainwater-harvesting umbrella prototype addressing urban water conservation through sustainable product design.
- **Institute-Level Finalist, Smart India Hackathon 2025** - Led the ideation and development of **FinSecure AI**.
- **Selected for UNESCO-ECF** Values Education Programme: Completed a month-long leadership programme and delivered a capstone presentation on UNESCO values and global citizenship.
- **Acharya Tulsi Gold Medal** (2024), **Shiksha Gaurav Samman** (2022), **10,000 prize** for academic excellence.

POSITIONS OF RESPONSIBILITY

Head of Speaker Relations & Marketing, TEDx IIT Hyderabad

- **Leading a 50+ member cross-functional team** across Speaker Relations and Marketing, driving strategic planning, operations, cross-domain collaboration, and execution for the institute's flagship **TEDx** conference.
- **Spearheading speaker acquisition and partnership strategy**, leading outreach, negotiations, and stakeholder management with industry leaders, startups, organizations, and institutional partners.
- **Driving marketing campaigns, strategic collaborations, and institute-wide event execution**, partnering with **NGOs, blind schools, educational institutions, and corporate organizations** to expand outreach, strengthen community engagement, and maximize event visibility.

PR & Networking Coordinator, TEDx IIT Hyderabad

- Expanded TEDx IIT Hyderabad's Instagram community by **500+ followers** through digital campaigns; led **speaker outreach and coordination as POC**, while organizing **X-MUN for 200+ participants** and institute-wide events.

Core Positions

- Vibes (Music Club, IITH) | RangDeManch (Theatre Arts Club, IITH) | Corporate Relations & Finance, E-Cell IITH

EXTRACURRICULAR

- Under-17 **District-level Football Player**, represented Jodhpur district in competitive football tournaments.
- **Represented IIT Hyderabad** at the **Inter IIT Cultural Meet (2024, 2025)** in the Theatre Arts Cup.
- **Winner, Inter-School Debate Competition** (2022) | **Winner, Battle of Bands Competition**, Diesta 2025, IIT-H
- **Gold Medalist, Vabri Math** (Abacus & Mental Mathematics competition).